

Where Does the Noise Come From? Understanding Challenges in Enhancing the Accuracy of Learning-Based Emotion Inference for Virtual Reality Applications Using EEG Signals

Yan-Ming Chen
National Tsing Hua University
Hsinchu, Taiwan
gepz@gapp.nthu.edu.tw

Amanda Castellanos
National Tsing Hua University
Hsinchu, Taiwan
s112065426@m112.nthu.edu.tw

Chih-Ching Chuang
National Tsing Hua University
Hsinchu, Taiwan
ch3@gapp.nthu.edu.tw

Yi-Chao Chen
Shanghai Jiao Tong University
Shanghai, China
yichao@sjtu.edu.cn

Chuang-Wen You
National Tsing Hua University
Hsinchu, Taiwan
cwyou@mx.nthu.edu.tw

Abstract

This study explored strategies to improve the quality of data collected in the training of emotion detection algorithms. A binary classification algorithm was developed to classify emotions as high or low in valence and arousal, based on EEG data recorded while participants viewed 360-degree Virtual Reality (VR) videos. Participants also self-reported their emotional responses through a VR-based interface, providing ground-truth labels for model training. Through a combination of quantitative and qualitative observations, we identified several challenges in the data collection process, including issues related to user discomfort, EEG signal consistency, and the reliability of self-reported emotions. In response, we proposed practical solutions to improve emotion inference accuracy by minimizing VR-induced discomfort, enhancing EEG data quality, and standardizing self-report procedures.

CCS Concepts

• Human-centered computing → Empirical studies in HCI.

Keywords

Emotion Inference, Neurofeedback, Virtual Reality.

ACM Reference Format:

Yan-Ming Chen, Amanda Castellanos, Chih-Ching Chuang, Yi-Chao Chen, and Chuang-Wen You. 2025. Where Does the Noise Come From? Understanding Challenges in Enhancing the Accuracy of Learning-Based Emotion Inference for Virtual Reality Applications Using EEG Signals. In *Companion of the 2025 ACM International Joint Conference on Pervasive and Ubiquitous Computing (UbiComp Companion '25)*, October 12–16, 2025, Espoo, Finland. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3714394.3754354>

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.

UbiComp Companion '25, Espoo, Finland

© 2025 Copyright held by the owner/author(s). Publication rights licensed to ACM.
ACM ISBN 979-8-4007-1477-1/2025/10
<https://doi.org/10.1145/3714394.3754354>

1 Introduction

The ability to detect and identify emotional states is of vital importance in psychological counseling, marketing, and human-computer interactions [8, 25, 37]. Emotion detection methods are increasingly being integrated into virtual reality (VR) systems to enhance the immersive quality of virtual meetings, training simulations, and social interactions [2, 26, 31, 40].

Publicly available EEG/ECG datasets—such as DEAP [19] and DREAMER [18]—have fueled progress in the use of deep learning for valence and arousal classification [36]; however, a reliance on 2D videos has constrained the intensity and nuance of emotional responses. Nonetheless, the use of EEG data for valence and arousal classification in immersive VR remains underexplored, while headset discomfort and EEG signal artifacts have hindered the acquisition of clean neural data in VR settings [6]. This exploratory study sought to refine EEG-based data collection protocols for immersive VR environments.

A pilot study involving 12 participants examined the effects of individual differences in EEG signals and self-reports on emotion-inference accuracy. Analysis of quantitative and qualitative observations revealed three primary challenges: (1) challenges and diversity in VR scene perception and comprehension, (2) inconsistent self-reports reflecting overlapping or ambiguous emotional states, and (3) disruptions in user focus caused by physical discomfort from EEG and VR hardware. In response, we identified several key observations, including the need to simplify scene design for VR videos, opportunities to leverage sensors to identify emotionally salient video segments in VR, and the necessity of ergonomic modifications to EEG devices and headsets. These insights can inform future refinements aimed at enhancing both data quality and participant experience, ultimately contributing to the advancement of reliable EEG-based emotion classification models for immersive VR applications.

2 Pilot Study: Inferring emotions in VR

2.1 Participants

Snowball sampling was used to recruit 12 participants (E01 to E12) from National Tsing Hua University. The participants were a mix

of undergraduate and graduate students, including one male and eleven females, aged 18 to 28 years (mean=22.58, SD=3.00).

2.2 Device

Emotional responses were elicited by immersing users in 360-degree video clips using a Meta Quest Pro VR headset. Throughout the presentation, EEG brain signals were recorded in real-time using the Emotiv EPOC X system [10] with 14 electrodes positioned over the left-brain hemisphere (AF3, F3, FC5, F7, O1, and T7) and right-brain hemisphere (AF4, F4, FC6, F8, O2, and T8). EEG signals were analyzed using EmotivPro software to derive the spectral power across the alpha, beta, gamma, theta, and delta frequency bands. Spectral and statistical analyses were conducted using Python libraries.

2.3 360-Degree VR Video Clips

This study utilized a public database of 360-degree VR videos [23], each of which had previously been evaluated in terms of arousal level based on the Self-Assessment Manikins (SAM) scale [4]. As proposed by Russell, twelve test clips were selected to ensure an even distribution across the valence-arousal emotional space [20, 32, 39]. A minimum duration of 60 seconds is generally recommended to elicit noticeable emotional changes [41, 42]. Accordingly, all videos were trimmed to a uniform length of 90 seconds with final 30 seconds used as input samples for predicting emotional response.

2.4 Binary Classification of Valence and Arousal

Pre-processing with a robust median filter [3] was used to mitigate noise-related spikes in the raw EEG signals [28], while preserving important signal characteristics. A Butterworth bandpass filter (0.5 – 50 Hz) was then applied to isolate frequencies relevant to neural activity [12]. From the last 30 seconds of filtered baseline data (i.e., collected before watching any VR videos), we computed the mean and standard deviation for use in normalizing the filtered data, ensuring values primarily between the range of -1 and 1.

Various features in the normalized EEG data were used to extract meaningful information from EEG signals captured during the last 30 seconds of each 90-second VR video. A 5-second sliding window with 50% overlap was used to segment the data for fine-grained feature extraction. In line with protocols established for the learning-based inference of emotions from EEG signals, this study employed three feature sets: time-domain, frequency-domain, and entropy-based. The 112 time-domain features covered a variety of metrics, including mean, standard deviation, peak-to-peak value, root mean square, zero-crossing numbers, and Hjorth parameters [13]. The 364 frequency-domain features included the total energy across five frequency bands (alpha, beta, gamma, theta, and delta), band power and average power within each band, and differences in power between specific electrode pairs to assess hemispheric asymmetry (e.g., AF3-AF4, F3-F4, FC5-FC6, F7-F8, O1-O2, and T7-T8). Additional metrics in the frequency-domain included the IMF (Intrinsic Mode Function) energy [38], wavelet energy [30], and power ratios for selected electrode pairs. The 224 entropy-based features comprised spectral entropy for each band as well as the differential entropy [29] of spectral power, IMF energy, and wavelet energy within each band. The resulting 700 features from each

5-second normalized EEG data segment provided a comprehensive representation of characteristics applicable to emotion inference.

The feature selection process was based on Jensen–Shannon (JS) distance [11], which quantifies the “distance” between two probability distributions to define a score for use in measuring the discriminative power of specific features. This was achieved by dividing the samples into groups of near equal size, including two groups with a ground truth label of high valence (HV_1 and HV_2) and two groups with a ground truth label of low valence (LV_1 and LV_2). The distance between any two groups of data was calculated using the JS distance ($JS(G_i, G_j)$). The distance between groups sharing the same ground truth label was minimized to ensure similar distributions, while the distance between groups with different ground truth labels was maximized to ensure distinct distributions. This approach was intended to facilitate the differentiation of high and low valence labels. The differentiability ($DA(i)$) of the i_{th} feature was defined as follows: $DA(i) = JS_{diff}(i) / JS_{same}(i)$. The numerator $JS_{diff}(i)$ represents the average Jensen–Shannon (JS) distance between all group-pairs with different ground truth labels ($\{HV_1, LV_1\}$, $\{HV_2, LV_2\}$, $\{HV_1, LV_2\}$, and $\{HV_2, LV_1\}$), indicating the efficacy of a given feature in differentiating among different ground truth labels. The denominator indicates the average JS distance between groups with the same ground truth label ($\{HV_1, HV_2\}$, and $\{LV_1, LV_2\}$), ensuring consistency within labels. Higher $DA(i)$ values indicate that a given feature is more effective at differentiating between high and low valence groups. The features were subsequently ranked according to $DA(i)$ values. Principal Component Analysis (PCA) [1] was then applied to the ranked feature set to reduce dimensionality, retaining principal components that collectively explained 98% of the total variance. After identifying key components, projected PCA features were input into either XGBoost [7] or Random Forest [9] algorithms to classify each sample as either high or low valence (HV/LV). The same procedure was used to classify data as high arousal (HA) or low arousal (LA).

2.5 Procedure

The researchers first outlined the objectives of the pilot study and then assisted the participants in donning the VR headset and EEG sensing device. The researchers familiarized participants with the use of the VR interface to navigate through the simulation and submit self-reports on their arousal levels based on the Self-Assessment Manikin (SAM) scale [4] using intuitive point-and-click gestures.

During the first 90 seconds of the session, the participants followed VR instructions aimed at inducing a state of relaxation, during which baseline EEG data were recorded for subsequent analysis. The participant subsequently viewed 8 videos, randomly chosen from a pool of 12 and shown in a random sequence. Ground-truth valence and arousal levels were obtained through the VR interface using the SAM scale (range: 1 (low) to 9 (high)) immediately after each video. Participants were also asked to follow the Think-Aloud method [22], verbalizing the elements that influenced their valence and arousal ratings. The same procedure was repeated for all eight test clips, with the presentation order counterbalanced across participants to minimize order effects. Emotional states were assessed immediately after each video to capture emotional responses.

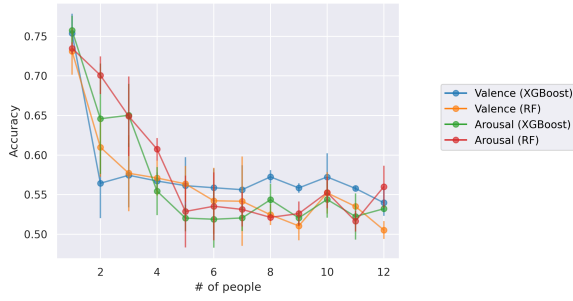


Figure 1: Accuracy of predictions for valence and arousal using the XGBoost or Random Forest algorithms.

After viewing eight VR videos, a post-study semi-structured interview was conducted to gain deeper insights into perceived emotions of participants. The interview focused on three main areas: (1) how participants perceived and interpreted the VR content and how these interpretations influenced their emotional responses; (2) difficulties encountered while self-reporting their induced emotions for the videos, along with feedback on the design and utility of the self-report interface and methods; and (3) any issues affecting their emotional state while watching the VR videos.

2.6 Data Analysis

Average prediction accuracy was calculated as the mean accuracy across a three-times-repeated four-fold cross-validation. Ranking of the i -th feature was determined according to the value of $JS(i)$.

Audio recordings of the semi-structured interviews were manually transcribed to identify key themes. Qualitative data, including transcripts obtained during the think-aloud process and post-study interviews, underwent iterative coding by at least two experienced researchers working independently. Codes with a reasonable level of agreement among the coders were used to derive salient themes via thematic analysis [5].

2.7 Results

2.7.1 Quantitative Results. Figure 1 illustrates the average accuracy in the binary classification of valence and arousal using the XGBoost and Random Forest algorithms. We first calculated the average accuracy based on data from a single participant (E01). The initial accuracy obtained using the Random Forest was as follows: valence (0.73) and arousal (0.73). The initial accuracy obtained using XGBoost was as follows: valence (0.75) and arousal (0.76). The average accuracy progressively declined as data from other participants was incrementally added. We then calculated the average accuracy for all 12 participants. The accuracy achieved using the Random Forest was as follows: valence (0.51) and arousal (0.56). The accuracy achieved using XGBoost was as follows: valence (0.54) and arousal (0.53). This trend highlights the challenges in maintaining classification accuracy as the diversity of training data increases.

Table 1 lists the top-3 features for all participants ranked according to the $JS(i)$ scores calculated for E01 to clarify the reasons underlying the observed decrease in classification performance.

This table also lists the rankings of the various features when evaluated using data from other participants. This analysis revealed that the top-ranked features for E01 did not consistently achieve similarly high rankings for the other participants. In other words, the top-3 rankings varied significantly across participants. In fact, many features received high rankings from some participants and low rankings from others. This inconsistency suggests underlying factors influencing feature performance.

Based on the average values for most of the videos, the quadrants of the valence-arousal space matched the quadrants reported in the literature [23]. The one exception was the second video (Tomorrowland 2014), which presented a significantly lower average valence score of 4.40. We also observed variations in the standard deviation of valence and arousal scores for some of the videos in a given label group. For instance, the 11th video (The Displaced) presented a higher standard deviation of valence (1.67), whereas the 10th video (War Zone) presented a higher standard deviation of arousal (1.20), compared to the other two videos in the LVLA category from the pre-evaluated results [23].

2.7.2 Qualitative Results

2.7.2.1 Challenges and Diversity in VR Scene Perception and Comprehension. Some participants reported that scene changes within a given video introduced elements that were not intuitively connected to the previous scenes. In some situations, the confusion induced by this lack of continuity hindered the evocation of the intended emotion. After watching the Zombie Apocalypse Horror video, E06 made the following comment: “At first, I was feeling a little nervous, but after the scene switch, I became confused... By the end, I wasn’t at all sure what was going on.” The inclusion of special effects in a video was also shown to influence the way that the participants perceived the evoked emotions. For instance, the fast-forward effect in the Sunset of Oia-Santorini video appeared confusing to at least one participant. E12 made the following comment: “I wasn’t sure what it was supposed to be... At first it felt calming, but the fast-forward effect made me go ‘Huh?’” . . . Also, I found it a bit boring. I didn’t dislike it, but I was not comfortable either. Could I rate this arousal as neutral?”

2.7.2.2 Challenges in Self-Reporting Ground Truth Emotions. Participants reported that their emotional responses changed as the videos progressed. It appears that the emotional response is strongly associated with the degree to which elements in the video relate to real-life experiences. E03 made the following comment: “The opening scene of people arguing created a heavy and intense atmosphere. However, as the story progressed, it began to feel like a theatrical performance – kind of like a zombie in a theater club. The guy next to me was laughing really hard, which I found both amusing and slightly awkward. My initial feeling of nervousness gradually faded to be replaced with a feeling of humorous intrigue.” Some participants reported that the videos triggered in them a vague desire to visit the places they were shown, rather than a specific emotion. In a few cases, their musings cause them to forget to self-report their emotions. E04 made the following comment: “After watching ‘Pacific Sunset Half Moon Bay’, I felt very calm – so I gave it valence score of 9. As for arousal... this video made me feel anxious to take action and go to the beach. [But didn’t you just say you felt very calm?]”

Table 1: Feature rankings based on $DA(i)$: Rankings for each participant and the top-3 features for E01. The naming convention for time-domain features is structured as (electrode name)-(statistical property). The naming convention for frequency- and entropy-domain features follows the format (electrode name)-(frequency band)-(statistical property).

Feature	E01	E02	E03	E04	E05	E06	E07	E08	E09	E10	E11	E12	Avg. (Std)
F3-Alpha-IMF Energy	1	498	435	299	323	290	380	213	26	699	75	316	296.25 (192.88)
F3-Alpha-Wavelet Entropy	2	694	643	512	232	578	524	570	1	178	696	690	443.33 (254.67)
FC5-Delta-Wavelet Entropy	3	226	62	445	425	473	517	692	657	123	584	444	387.58 (220.94)

Oh... I just realized that my eagerness to go to the beach seems to be more of a future plan I thought about when you asked me during the post-study interview. While watching the video, I was indeed calm."

2.7.2.3 Influence of Wearable Devices on VR Immersion and Emotional Perception. Some of the VR videos were presented from the first-person perspective, such that participants felt fully immersed in their experience of the VR world. It appears that this connectedness enhanced their emotional engagement with the story. Other videos were presented from a bystander's perspective, as though observing events unrelated to themselves. Their lack of connectedness appears to have reduced their emotional engagement. E01 made the following comment: *"The first video [Jailbreak 360] is shown from the perspective of a bystander and the event has already happened. But this one [War Zone] employed a first-person perspective, making me feel as if I were part of the war, which heightened my emotional engagement."* Despite the fact that the VR device and EEG devices were quite light, the participants complained that they became uncomfortable over time. In fact, the weight and imperfect fit actually interfered with the viewing experience of some participants. E08 made the following comment: *"The VR headset kept pressing parts of my head, causing discomfort. [...] It also continuously pressed against my glasses, which put pressure on the bridge of my nose."*

3 Discussions

3.1 Simplify the Scene Design for VR Videos

Developers should avoid making video clips that include multiple scenes with distinct visual elements and/or angles of view, each of which could potentially elicit a different emotional response [15]. Viewers need time to fully comprehend the narrative and the intended aim of the video. An overly complex storyline or location could confuse the viewer, thereby hindering the perception of elements meant to induce an emotional response, particularly when the scene duration is short. Undue complexity also increases the risk of inducing unintended emotions. Likewise, special effects should be used sparingly to avoid the wide range of emotional responses that these devices could elicit among different users.

3.2 Using Sensors to Facilitate the Identification of Emotional Video Segments in VR

As mentioned above, every visual element in every scene could potentially elicit a different emotional response. Nonetheless, many stories cannot be conveyed without a complex setting. When dealing with complex scenarios, the self-report user interface should be altered to allow participants to navigate through the video to rate their emotions scene-by-scene. The VR interface should also

include a feature that enables participants to highlight emotionally charged segments or specific time periods associated with powerful emotional responses (via a gesture or input device). We also advocate the inclusion of physiological sensors capable of detecting salient emotional responses [43, 45], thereby eliminating the need for participants to continuously self-report their emotions [46]. Sensor-assisted detection could be used as a complement to manual input to further enhance annotation. We are confident that these simple changes would result in a more nuanced report and enhance the quality of self-reported ground truth data.

3.3 Enhancing the Usability of EEG and VR Devices

Subjects wearing both EEG and VR devices are prone to discomfort, irritation, and a lack of focus, necessitating careful management of session durations. Each session in this study was limited to 8 videos (approximately 30 minutes), which unexpectedly caused discomfort in some participants. Future research should focus on shorter sessions with rest intervals between them. Researchers should also explore ergonomic solutions, such as those used in high-end headphones, to minimize device-related discomfort.

EEG has outstanding potential for the identification of valence and arousal states; however, current classification methods can be confounded by variations in sensing hardware, seriously impeding the training of deep learning algorithms. Electrode [16, 33, 44], signal noise [21, 34], and differences at the subject level [27, 35] often result in inconsistent measurements, as evidenced by reported differences in feature rankings (see Section 2.7.1). In the future, researchers should standardize the implementation of EEG devices across experimental sessions. Advanced processing methods should also be considered to enhance feature extraction [17, 21], improve EEG signal quality, and remove artifacts [14, 24]. Improving electrode placement and minimizing signal noise could have a profound effect on the accuracy and reliability of EEG-based measurements.

4 Conclusion

This pilot study combined quantitative and qualitative analyses to pinpoint challenges in EEG data collection for VR-based emotion detection. We identified key factors—such as inconsistent EEG signals, unreliable self-reports, and user discomfort—that undermine classification accuracy and discussed directions of refinements.

Acknowledgments

This research was supported by the National Science and Technology Council of Taiwan (NSTC 113-2221-E-007-110 and 114-2221-E-007-077-MY2).

References

- [1] Hervé Abdi and Lynne J. Williams. 2010. Principal component analysis. *WIREs Computational Statistics* 2, 4 (2010), 433–459. arXiv:https://wires.onlinelibrary.wiley.com/doi/pdf/10.1002/wics.101 doi:10.1002/wics.101
- [2] Mesut Akdere, Yeling Jiang, and Flavio Destri Lobo. 2021. Evaluation and assessment of virtual reality-based simulated training: exploring the human–technology frontier. *European Journal of Training and Development* 46, 5/6 (2021), 434–449.
- [3] Nima Bigdely-Shamlo, Tim Mullen, Christian Kothe, Kyung-Min Su, and Kay A. Robbins. 2015. The PREP pipeline: standardized preprocessing for large-scale EEG analysis. *Frontiers in Neuroinformatics* 9 (2015). doi:10.3389/fninf.2015.00016
- [4] Margaret M. Bradley and Peter J. Lang. 1994. Measuring emotion: The self-assessment manikin and the semantic differential. *Journal of Behavior Therapy and Experimental Psychiatry* 25, 1 (1994), 49–59. doi:10.1016/0005-7916(94)90063-9
- [5] Virginia Braun, Victoria Clarke, Nikki Hayfield, and Gareth Terry. 2019. Thematic Analysis. In *Handbook of Research Methods in Health Social Sciences*, Praneet Liampattong (Ed.). Springer Singapore, 843–860.
- [6] Mila Buić, Mikko Salminen, and Juho Hamari. 2023. Effects of Immersive Media on Emotion and Memory: An Experiment Comparing Article, 360-video, and Virtual Reality. *International Journal of Human-Computer Studies* 179 (2023), 103118. doi:10.1016/j.ijhcs.2023.103118
- [7] Tianqi Chen and Carlos Guestrin. 2016. XGBoost: A Scalable Tree Boosting System. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining* (San Francisco, California, USA) (KDD '16). Association for Computing Machinery, New York, NY, USA, 785–794. doi:10.1145/2939672.2939785
- [8] R. Cowie, E. Douglas-Cowie, N. Tsapatsoulis, G. Votsis, S. Kollias, W. Fellenz, and J.G. Taylor. 2001. Emotion recognition in human-computer interaction. *IEEE Signal Processing Magazine* 18, 1 (2001), 32–80. doi:10.1109/79.911197
- [9] Adele Cutler, D. Richard Cutler, and John R. Stevens. 2012. *Random Forests*. Springer New York, New York, NY, 157–175. doi:10.1007/978-1-4419-9326-7_5
- [10] EMOTIV. 2025. Emotiv EPOC X. <https://www.emotiv.com/epoc-x/>.
- [11] D.M. Endres and J.E. Schindelin. 2003. A new metric for probability distributions. *IEEE Transactions on Information Theory* 49, 7 (2003), 1858–1860. doi:10.1109/TIT.2003.813506
- [12] Ishita Goyal and Anmol Mehta. 2021. Acquisition, Pre-Processing, and Feature Extraction of EEG Signals to Convert it into an Image Classification Problem. <https://api.semanticscholar.org/CorpusID:236454504>
- [13] Bo Hjorth. 1970. EEG analysis based on time domain properties. *Electroencephalography and clinical neurophysiology* 29, 3 (1970), 306–310.
- [14] Li Hu and Zhiguo Zhang. 2019. *EEG signal processing and feature extraction*. Springer.
- [15] Xiansheng Hua and Hongjiang Zhang. 2004. Content and transformation effect matching for automated home video editing. *2004 International Conference on Image Processing, 2004. ICIP '04*. 3 (2004), 1613–1616. <https://api.semanticscholar.org/CorpusID:9761122>
- [16] Herbert H Jasper. 1958. Ten-twenty electrode system of the international federation. *Electroencephalogr Clin Neurophysiol* 10 (1958), 371–375.
- [17] Xiao Jiang, Gui-Bin Bian, and Zean Tian. 2019. Removal of artifacts from EEG signals: a review. *Sensors* 19, 5 (2019), 987.
- [18] Stamos Katsigiannis and Naeem Ramzan. 2018. DREAMER: A Database for Emotion Recognition Through EEG and ECG Signals From Wireless Low-cost Off-the-Shelf Devices. *IEEE Journal of Biomedical and Health Informatics* 22, 1 (2018), 98–107. doi:10.1109/JBHI.2017.2688239
- [19] Sander Koelstra, Christian Muhl, Mohammad Soleymani, Jong-Seok Lee, Ashkan Yazdani, Touradj Ebrahimi, Thierry Pun, Anton Nijholt, and Ioannis Patras. 2012. DEAP: A Database for Emotion Analysis Using Physiological Signals. *IEEE Transactions on Affective Computing* 3, 1 (2012), 18–31. doi:10.1109/T-AFFC.2011.15
- [20] Peter Kuppens, Francis Tuerlinckx, Michelle Yik, Peter Koval, Joachim Coosemans, Kevin J Zeng, and James A Russell. 2017. The relation between valence and arousal in subjective experience varies with personality and culture. *Journal of personality* 85, 4 (2017), 530–542.
- [21] Chi Qin Lai, Haidi Ibrahim, Mohd Zaid Abdullah, Jafri Malin Abdullah, Shahrel Azmin Suandi, and Azlinda Azman. 2018. Artifacts and noise removal for electroencephalogram (EEG): A literature review. In *2018 IEEE Symposium on Computer Applications & Industrial Electronics (ISCAIE)*. IEEE, 326–332.
- [22] C. Lewis. [n. d.]. *Using the "thinking Aloud" Method in Cognitive Interface Design*. IBM Thomas J. Watson Research Division. <https://books.google.com.tw/books?id=F5AKHQAACAJ>
- [23] Benjamin J Li, Jeremy N Bailenson, Adam Pines, Walter J Greenleaf, and Leanne M Williams. 2017. A public database of immersive VR videos with corresponding ratings of arousal, valence, and correlations between head movements and self report measures. *Frontiers in psychology* 8 (2017), 2116.
- [24] Fabien Lotte and Marco Congedo. 2016. EEG feature extraction. *Brain–Computer Interfaces 1: Foundations and Methods* (2016), 127–143.
- [25] Paulo PP Machado, Larry E Beutler, and Leslie S Greenberg. 1999. Emotion recognition in psychotherapy: Impact of therapist level of experience and emotional awareness. *Journal of Clinical Psychology* 55, 1 (1999), 39–57.
- [26] Javier Marin-Morales, Carmen Llinares, Jaime Guixeres, and Mariano Alcañiz. 2020. Emotion recognition in immersive virtual reality: From statistics to affective computing. *Sensors* 20, 18 (2020), 5163.
- [27] Gerald Matthews and Manfred Amelang. 1993. Extraversion, arousal theory and performance: A study of individual differences in the EEG. *Personality and individual differences* 14, 2 (1993), 347–363.
- [28] Umberto Melia, Francesc Clariá, Montserrat Vallverdú, and Pere Caminal. 2014. Filtering and thresholding the analytic signal envelope in order to improve peak and spike noise reduction in EEG signals. *Medical Engineering & Physics* 36, 4 (2014), 547–553. doi:10.1016/j.medengphy.2013.11.014
- [29] Joseph Victor Michalowicz, Jonathan M Nichols, and Frank Bucholtz. 2013. *Handbook of differential entropy*. Crc Press.
- [30] Zeynab Mohammadi, Javad Frounchi, and Mahmood Amiri. 2017. Wavelet-based emotion recognition system using EEG signal. *Neural Computing and Applications* 28 (2017), 1985–1990.
- [31] Seyed Muhammad Hossein Mousavi, Bulat Khaertdinov, Pedro Jeuris, Enrique Hortal, Davide Andreoletti, and Silvia Giordano. 2023. Emotion Recognition in Adaptive Virtual Reality Settings: Challenges and Opportunities. *WAMWB@ MobileHCI* (2023), 1–20.
- [32] David Olson, Candyce Smith Russell, and Douglas H Sprenkle. 2014. *Circumplex model: Systemic assessment and treatment of families*. Routledge.
- [33] Robert Oostenveld and Peter Praamstra. 2001. The five percent electrode system for high-resolution EEG and ERP measurements. *Clinical neurophysiology* 112, 4 (2001), 713–719.
- [34] Jan Pieter Pijn, Jan Van Neerven, Andre Noest, and Fernando H Lopes da Silva. 1991. Chaos or noise in EEG signals; dependence on state and brain site. *Electroencephalography and clinical Neurophysiology* 79, 5 (1991), 371–381.
- [35] John Polich. 1997. On the relationship between EEG and P300: individual differences, aging, and ultradian rhythms. *International journal of psychophysiology* 26, 1-3 (1997), 299–317.
- [36] Md. Mustafizur Rahman, Ajay Krishno Sarkar, Md. Amzad Hossain, Md. Selim Hossain, Md. Rabiul Islam, Md. Biplob Hossain, Julian M.W. Quinn, and Mohammad Ali Moni. 2021. Recognition of human emotions using EEG signals: A review. *Computers in Biology and Medicine* 136 (2021), 104696. doi:10.1016/j.compbmed.2021.104696
- [37] Bernardete Ribeiro, Gonçalo Oliveira, Ana Laranjeira, and Joel Arrais. 2017. Deep learning in digital marketing: brand detection and emotion recognition. *International Journal of Machine Intelligence and Sensory Signal Processing* 2 (01 2017), 32. doi:10.1504/IJMISSP.2017.088173
- [38] Gabriel Rilling, Patrick Flandrin, and Paulo Gonçalves. 2003. On empirical mode decomposition and its algorithms. <https://api.semanticscholar.org/CorpusID:16054541>
- [39] J.A. Russell. 1980. A circumplex model of affect. *Journal of personality and social psychology* 39, 6 (1980), 1161–1178.
- [40] Jannik Rößler, Jiachen Sun, and Peter Gloor. 2021. Reducing Videoconferencing Fatigue through Facial Emotion Recognition. *Future Internet* 13, 5 (2021). doi:10.3390/fi13050126
- [41] Alexandre Schaefer, Frédéric Nils, Xavier Sanchez, and Pierre Philippot. 2010. Assessing the effectiveness of a large database of emotion-eliciting films: A new tool for emotion researchers. *Cognition and emotion* 24, 7 (2010), 1153–1172.
- [42] Mohammad Soleymani, Sadjad Asghari-Esfeden, Yun Fu, and Maja Pantic. 2015. Analysis of EEG signals and facial expressions for continuous emotion detection. *IEEE Transactions on Affective Computing* 7, 1 (2015), 17–28.
- [43] Mohammad Soleymani, Joep J.M. Kierkels, Guillaume Chanel, and Thierry Pun. 2009. A Bayesian framework for video-to-affective representation. In *2009 3rd International Conference on Affective Computing and Intelligent Interaction and Workshops*. 1–7. doi:10.1109/ACII.2009.5349563
- [44] Michal Teplan et al. 2002. Fundamentals of EEG measurement. *Measurement science review* 2, 2 (2002), 1–11.
- [45] Ling Wang, Jiayu Hao, Tie Hua Zhou, and Tianshuo Bi. 2024. Heart Disease Prediction Model based on HRV Emotional Features Fusion. In *Proceedings of the 2024 8th International Conference on High Performance Compilation, Computing and Communications* (Guangzhou, China) (HP3C '24). Association for Computing Machinery, New York, NY, USA, 71–75. doi:10.1145/3675018.3675773
- [46] Tong Xue, Surjya Ghosh, Gangyi Ding, Abdallah El Ali, and Pablo Cesar. 2020. Designing Real-time, Continuous Emotion Annotation Techniques for 360° VR Videos. In *Extended Abstracts of the 2020 CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA) (CHI EA '20). Association for Computing Machinery, New York, NY, USA, 1–9. doi:10.1145/3334480.3382895